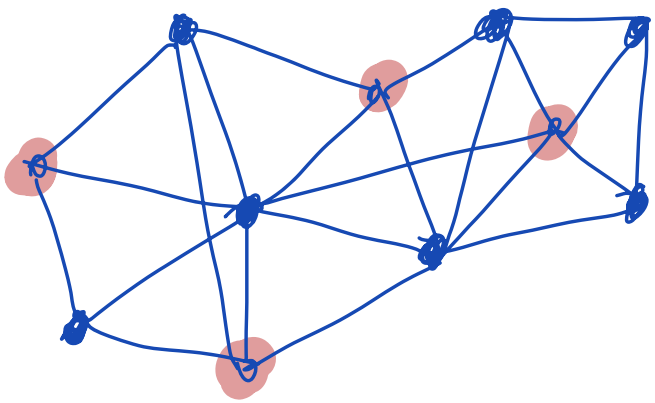
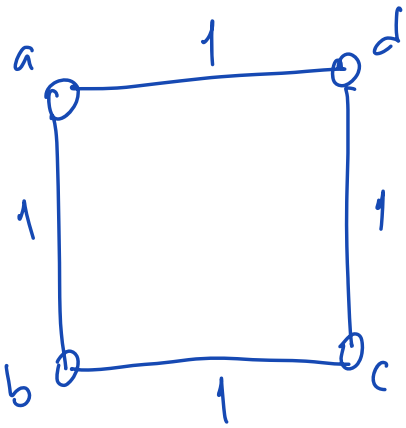
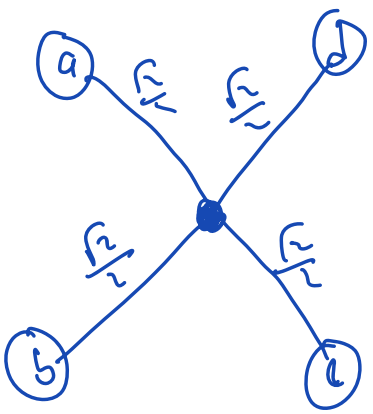
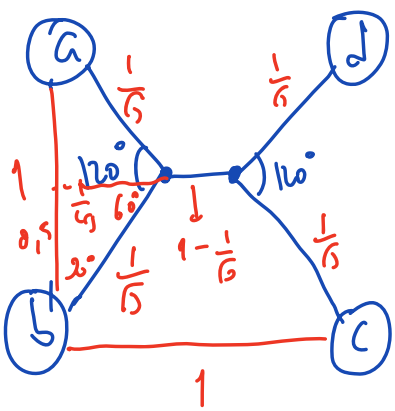




cost: 4



cost: $2\sqrt{2}$



$$\frac{4}{\sqrt{3}} + 1 - \frac{1}{\sqrt{3}} = 1 + \frac{3}{\sqrt{3}}$$

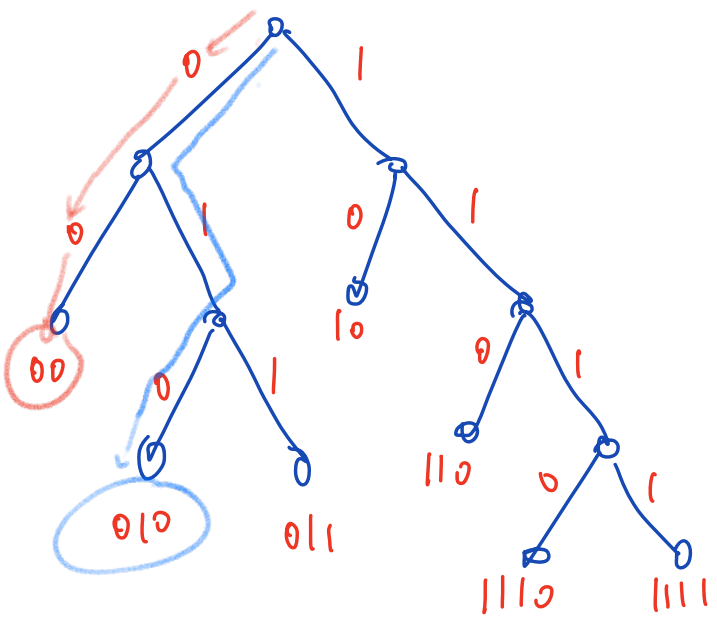
$$= 1 + \sqrt{3}$$

Coding Problems

A → 0010
B → 001
} not prefix-free.

B?
0 0 1 0 0 1
A?

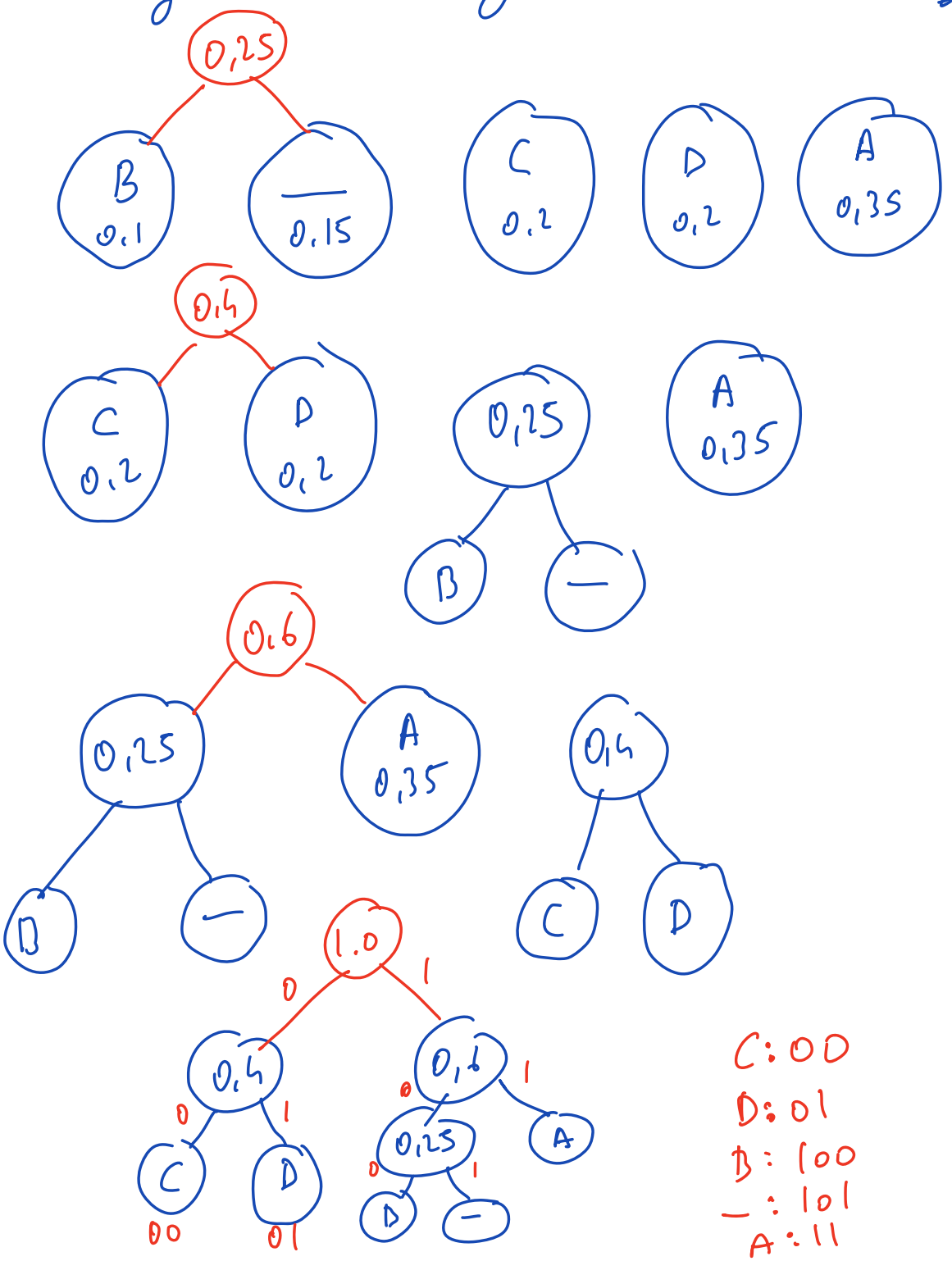
A → 0010
Z → 01
} not a good idea. A is more frequent than Z.



Codewords assigned to the leaf nodes are always prefix-free!

A: 0,35 B: 0,1 C: 0,2 D: 0,2 _ : 0,15

Let's say that we have only 5 characters with above frequencies.



Expected codeword length (weighted avg).

$$\underbrace{0,2 \times 2}_{C} + \underbrace{0,2 \times 2}_{D} + \underbrace{0,1 \times 3}_{B} + \underbrace{0,15 \times 3}_{-} + \underbrace{0,35 \times 2}_{A}$$

$$= 0,4 + 0,4 + 0,3 + 0,45 + 0,7 = 2,25$$

Best fixed-size encoding would require 3 bits.

$$2^2 < 5 \leq 2^3$$

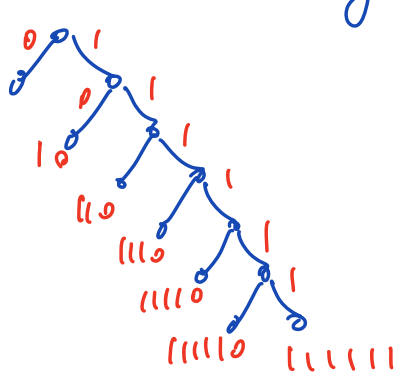
5 characters can be represented with 3 bits.

Therefore the compression rate w.r.t. the best fixed sized encoding:

$$\frac{3 - 2,25}{3} = 25\%$$

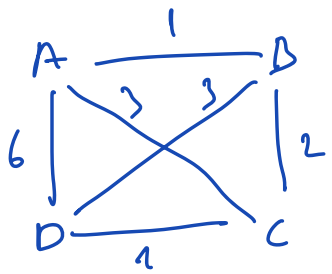
For n characters, what would be the largest possible codeword length?

$$\underline{\underline{n-1}}$$

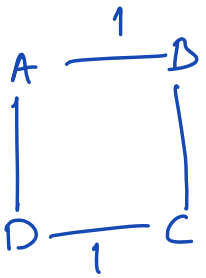


Greedy alg. for TSP

- nearest neighbor alg.
- Multi-fragment heuristic.



(A,B) 1 ✓
(D,C) 1 ✓
(B,C) 2 ✓
(A,C) 3 ✗
(B,D) 3 ✗
(A,D) 6 ✓



Greedy alg. for Knapsack Problem

		value/weight
$v_1 = 40$€	$w_1 = 2$ kg.	40/2 = 20
$v_2 = 30$€	$w_2 = 5$ kg	30/5 = 6
$v_3 = 50$€	$w_3 = 10$ kg	50/10 = 5
$v_4 = 10$€	$w_4 = 5$ kg.	10/5 = 2

$W = 16$

chose item 1 — value in the knapsack = 40€
remaining capacity = 16 − 2 = 14
chose item 2. rem. cap. = 14 − 5 = 9 — " " " " = 70€
cannot chose item 3.
chose item 4: rem. cap. 9 − 5 = 4 — " " " " = 80€

1 & 3 ⇒ 90° ⇒ better res. H.